

**What makes an
asset the
dominant
intermediary?**



**Centre for Blockchain
Technologies**

The Making of Dominant Vehicle Currencies

Evidence from DeFi
Jiahua Xu

How we reconstruct a route from one transaction

Direct swap

one transaction



Sequential vehicle route

one transaction



Parallel split

one transaction



Round trip

one transaction



Swap events in one transaction reveal sequential routes, parallel splits, and round trips.

Vehicle dominance is the share of indirect trade carried by an asset

MEASUREMENT

- Dominance is the share of indirect trade intermediated by asset k .
- Count share treats each route equally; value share weights routes by dollar value.
- Network measures capture how broadly k connects endpoint pairs.



Count-weighted dominance

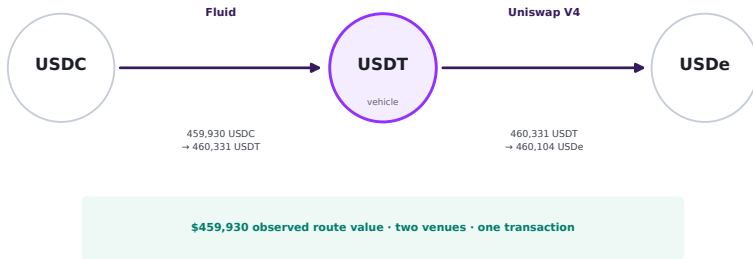
Share of indirect routes that pass through k

Value-weighted dominance

Share of comparable routed value that passes through k

This transaction routes USDC into USDe through USDT

OBSERVED ROUTE



The trace shows that USDT was used. It cannot tell us whether a direct USDC–USDe route was available or cheaper at that moment.

Architecture changes routing opportunities

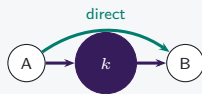
V1 mandatory hub



Only ETH-token pools: token-to-token execution must pass through ETH.

Hub use is imposed

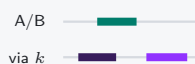
V2 pair choice



Arbitrary pairs add a direct route; vehicle routes remain feasible.

Hub use becomes a choice

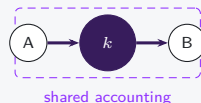
V3 range choice



LPs choose ranges and fees; active depth can favor direct pairs or vehicle spokes.

Opportunity shifts by pair

V4 net settlement



Two pools still execute; intermediate token movement may be netted.

Settlement and vehicle use can diverge

Architecture feasible rules

Exposure realised pool and route use

Calendar date timing only

The sample spans Uniswap V1 through V4

DATA



Routes are reconstructed transaction by transaction across the routed venue perimeter. Daily and weekly panels are derived from the same route units.

We measure vehicle dominance and distinguish it from demand, network position, and execution cost

1. **Vehicle share**

Fraction of route units intermediated by asset k .

2. **Excess use**

Intermediary share relative to endpoint demand.

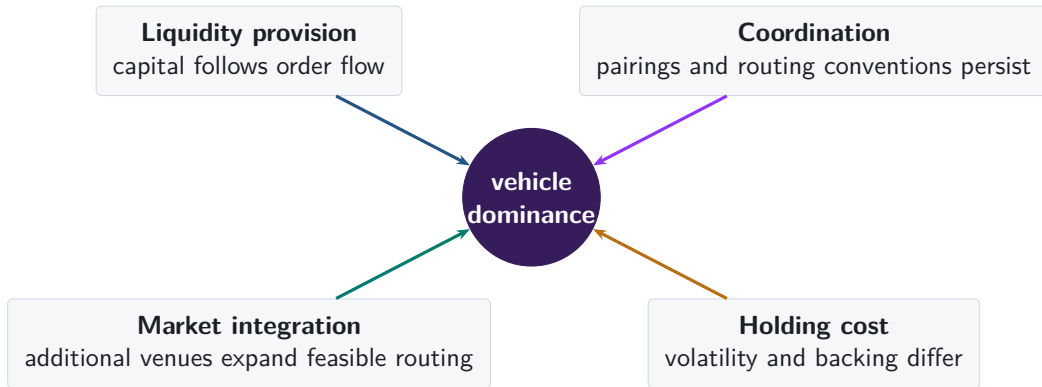
3. **Network position**

Degree, strength, and betweenness on the trading graph.

4. **Execution advantage**

Direct-route output relative to a feasible vehicle route at the same pre-trade state.

Why might one asset become the dominant vehicle?



We study how vehicle currencies emerge, gain share, and persist

FORMATION

Which rules make intermediation necessary or feasible?

Imposed intermediation, arbitrary pairing, concentrated liquidity, and singleton settlement.

ROTATION

Which assets gain share within comparable routes?

Count share, value share, excess use, network position, and route complexity over time.

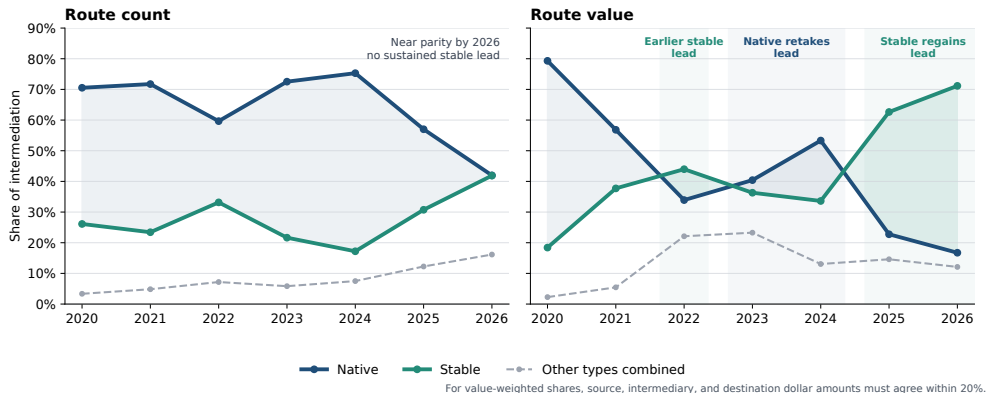
PERSISTENCE

Does an incumbent remain when a cheaper route is feasible?

Compare the incumbent with contemporaneous alternatives.

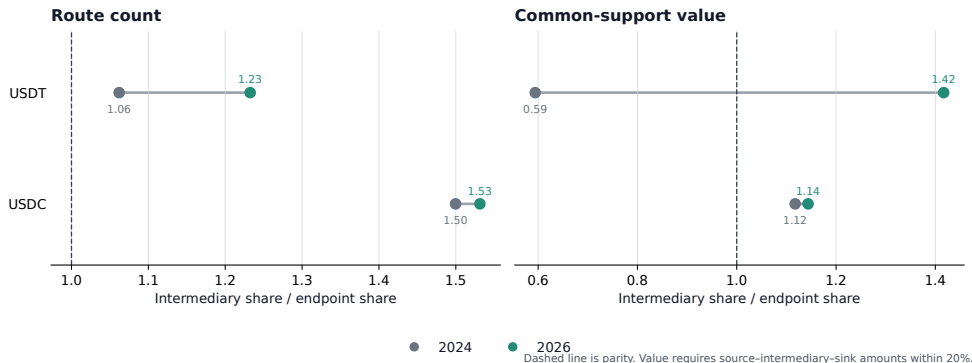


Stablecoins regain the lead in routed value



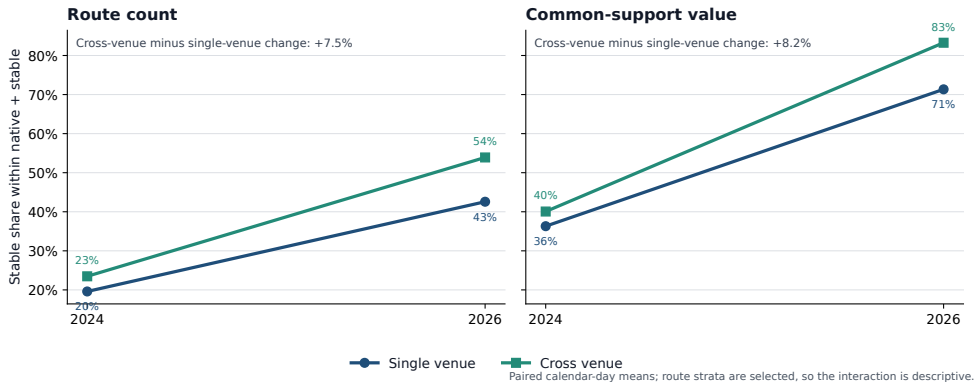
Annual shares of observed vehicle routes; the figure does not attribute the changes to protocol design.

USDT drives the post-2024 rise in stablecoin intermediation



“Common-support value” retains routes whose source, intermediary, and destination dollar values agree within 20%. A ratio above one means that the token appears more often as an intermediary than as an endpoint. USDT moves from below one to above one by routed value; USDC is above one in both years.

Stable share rises within both single- and cross-venue routes



Stablecoin intermediation rises in both groups, and more in cross-venue routes. Changing venue mix alone cannot explain the aggregate increase. Because route type is chosen, this comparison is descriptive.

Cheaper challengers gain share as incumbents lose it

30-day outcome	$\hat{\beta}$	SE	t
Challenger vehicle-share response	4.825	0.366	13.17
Incumbent vehicle-share response	-3.872	0.420	-9.22
Challenger overtakes incumbent	9.947	0.486	20.48

$N = 57,989$ pair-candidate observations; 2,213 date clusters. Exact alternating-projection absorption and CR1 rank correction.

The estimates are large in the small set of pair-candidate observations we can follow. They remain descriptive because the panel does not compare each pair against a fixed set of feasible routes.

Current architecture changes do not provide a clean before–after comparison

	5%	10%	25%
Sustained entries	162 (0)	139 (0)	78 (0)
Substitution exits	79 (0)	69 (0)	54 (0)
Role disappearances	7 (3)	7 (3)	7 (3)
Immediate entry contrast	–	–	–
Immediate exit contrast	–	–	–

At every threshold, entries and exits coincide with changes in the comparison set or with the appearance and disappearance of vehicle roles. These windows therefore cannot separate an architecture effect from changes in which pairs and vehicles remain active.

A before–after estimate requires the same feasible alternatives before and after the architecture change.

Stablecoins regain the value lead as route counts approach parity

- Stablecoins lead routed value in 2022 and again in 2025–2026, but fall behind native assets in 2023–2024.
- USDT drives the post-2024 rise in excess stablecoin intermediation; USDC changes little.
- The rise appears within both single- and cross-venue routes. Which mechanism drives it remains open.

Appendix map

Definitions and data

- A1 Route units and vehicle status
- A2 Asset types and backing regimes
- A3 Sample construction and exclusions

Reconstruction and validation

- A4 Exact transaction state
- A5 Protocol-specific state
- A6 Forced routes in V1

Robustness

- A7 Daily and weekly frequencies
- A8 Count and value weighting
- A9 Endpoint demand

Mechanisms

- A10 Capital, inventory, and depth
- A11 Coordination and integration
- A12–A13 References

A1. One reconstructed route is one unit



one route unit
two swap legs

- Each linked $i \rightarrow k \rightarrow o$ route is counted once.
- Asset k is classified as the intermediary on that route.
- Dominance is the share of route units or supported value carried by k .

A2. Backing regimes vary over time

Native platform asset

ETH, WETH; volatile settlement asset and protocol-native demand.

Fiat-reserve stablecoin

USDC, USDT; reserve and redemption design.

On-chain collateralised

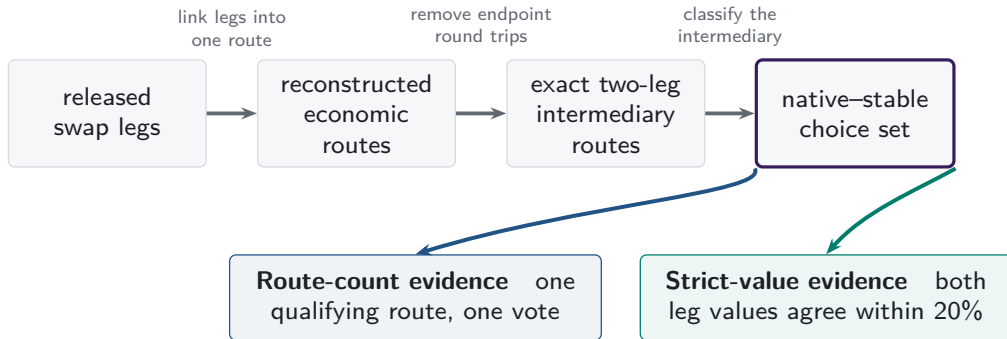
DAI, LUSD, crvUSD; collateral, liquidation, and backing changes.

Tokenised external asset

WBTC, tokenised gold; wrapped claim on an external asset.

Heterogeneity tests use each asset's contemporaneous backing regime.

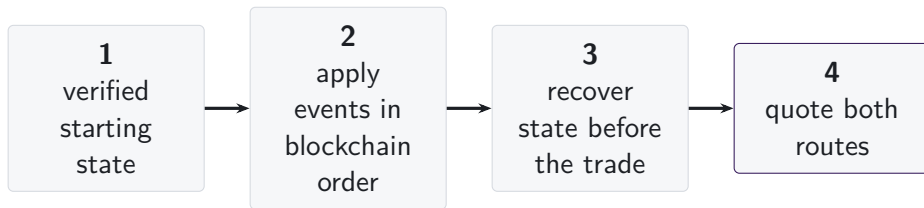
A3. One route universe supports two measurements



Topology-valid routes remain in count evidence when dollar support is unavailable.
Box widths do not represent attrition.

Missing dollar support narrows value-weighted evidence; it does not erase a topology-valid route from the count evidence.

A4. State is reconstructed immediately before execution



- Pool events are applied in blockchain order from a verified starting state.
- Each route is quoted using the pool state immediately before the transaction.
- Missing state intervals are excluded, and their share of trading volume is reported.

A5. Each AMM family has a different state vector

	reserves or active liquidity	ticks	amplification	weights or scaling	rates or oracles	fees	hooks
Constant product	■	—	—	—	—	■	—
Concentrated liquidity	■	■	—	—	—	■	—
StableSwap families	■	—	■	—	■	■	—
Weighted pools	■	—	—	■	■	■	—
V4 singleton	■	■	—	—	—	■	□

The common object is the state immediately before execution.
The fields required to recover that state vary with the pricing rule.

■ required state

□ hooks outside comparison

A6. V1 forced routes have an on-chain signature

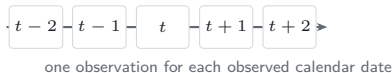
- Token-to-token execution emits two linked swaps through ETH.
- Matching ETH quantities distinguish protocol-imposed intermediation from unrelated bundled swaps.



same transaction, matching ETH flow

A7. Daily and weekly frequencies answer different questions

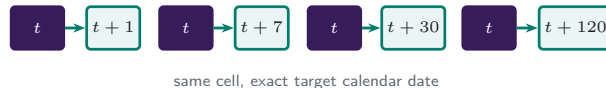
Daily panel



Complete weeks



Exact links



Model role

Daily panel date fixed effects

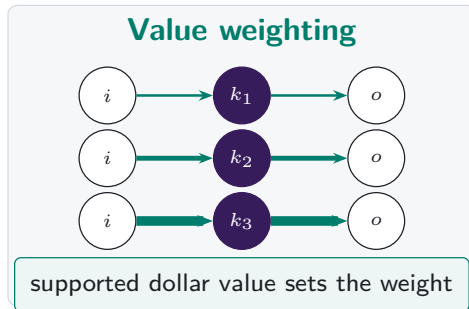
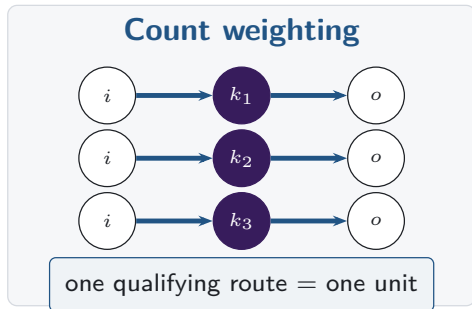
Calendar check week and weekday fixed effects

Complete weeks week fixed effects

Exact links inference by origin date

Missing target dates remain missing: row shifts and calendar months do not substitute for exact calendar links.

A8. Count and value define separate dominance margins



Both panels contain the same routes. Line widths are illustrative, and value weighting applies only on the stated valuation-support band.

A9. Endpoint demand and intermediary use are separate margins

Intermediary share

$$S_{k,t}^I = \frac{N_{k,t}^I}{\sum_j N_{j,t}^I}$$

How often asset k sits inside a route.

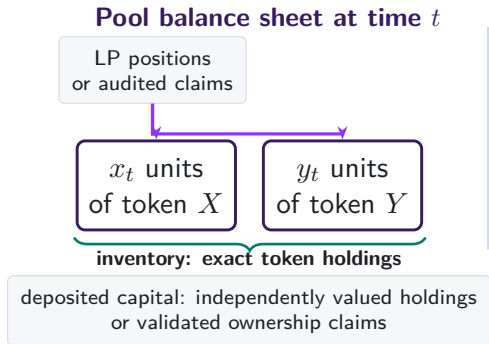
Endpoint share

$$S_{k,t}^E = \frac{N_{k,t}^{\text{in}} + N_{k,t}^{\text{out}}}{\sum_j N_{j,t}^{\text{in}} + N_{j,t}^{\text{out}}}$$

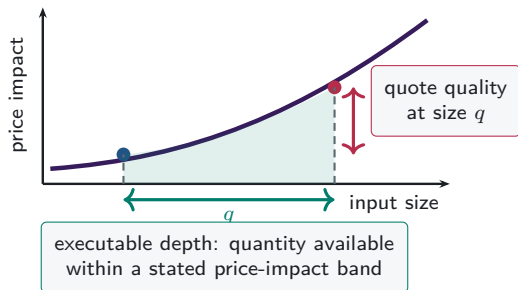
How often asset k is what traders buy or sell.

Excess use compares intermediation with endpoint demand on the same support.

A10. Capital, inventory, and depth are different objects



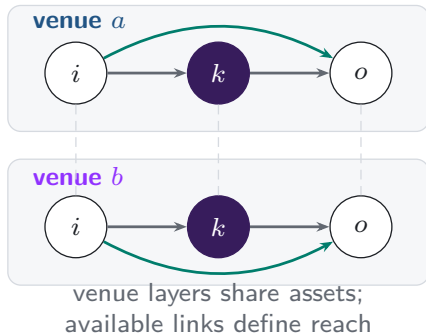
Execution from that state



Capital is a stock. Depth and quotes depend on trade size and current pool state.

A11. Integration changes the routes that can be chosen

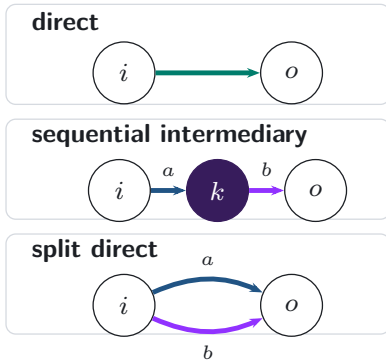
Feasible venue layers



realised
execution



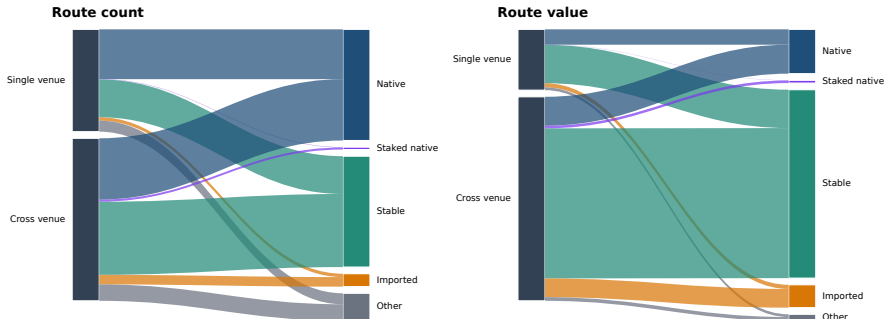
Realised route topology



The analysis therefore separates opportunity-set reach from the composition of realised routes.

A11b. Venue scope and vehicle type differ in 2026

Integration and vehicle composition are separate margins in 2026



Ribbon width is the share of 2026 intermediation jointly classified by venue scope and intermediary type.

The ribbons classify realised intermediation jointly by venue scope and intermediary type. They describe selected routes; they do not identify an effect of integration or the unchosen opportunity set.

A12. References: vehicle currencies and market structure

- Amiti, Itskhoki and Konings (2022), *Quarterly Journal of Economics*
- Dowd and Greenaway (1993), *Economic Journal*
- Flandreau and Jobst (2009), *Economic Journal*
- Gopinath and Stein (2021), *Quarterly Journal of Economics*
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A13. References: decentralised exchange design

- Adams et al. (2021), *Uniswap v3 Core*
- Angeris, Chitra, Evans and Boyd, optimal routing in constant-function markets
- Barbon and Ranaldo, decentralised exchange costs and market structure
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- Millionis, Moallemi, Roughgarden and Zhang, loss versus rebalancing
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